

Context-Aligned Reasoning Architecture (CARA)

A Structural Control Framework for Context-Sensitive AI Alignment and Mitigation of Absorption Drift

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Target Architecture: Stage A Modular Control Plane & Stage B Latent Co-processing

Abstract: Autonomous AI systems powered by large language and reasoning models increasingly demonstrate exceptional proficiency in multi-step task execution. However, long-horizon agency reveals a fundamental alignment vulnerability: *absorption drift*, wherein an agent reasons with high local competence while progressively detaching from the broader context, assumptions, and constraints that originally justified the task. We propose the **Context-Aligned Reasoning Architecture (CARA)**, a modular control and governance framework designed to maintain persistent contextual alignment. Rather than treating goal pursuit as an unconstrained optimization over static prompts, CARA formalizes goal validity as an actively evaluated invariant via an explicit **Goal–Context Binding** operator $B_t = f_{\theta}(G_t, C_t, V_t)$. We synthesize CARA into three operational planes—the State & Binding Plane, the Task Deliberation Plane, and the Governance & Safety Plane—and introduce an asynchronous boundary-gating protocol that separates deliberative reasoning from execution authority. Finally, we formulate a three-condition empirical benchmark protocol to measure absorption drift, step-to-detection latency, and false-interruption rates.

1. Introduction & The Problem of Absorption Drift

Modern reasoning agents excel at problem decomposition, tool use, and recursive sub-task execution. However, during extended execution trajectories, in-context learning and autoregressive attention distributions become saturated with intermediate artifacts, execution outputs, and local subproblems. This induces **absorption drift**:

- The model's attention is consumed by local task progress.
- Initial assumptions, stakeholder constraints, and contextual preconditions suffer from representational dilution.
- When environmental shifts invalidate the original objective, the model persists in execution because task competence remains intact even as task justification has evaporated.

CARA addresses this structural vulnerability not by prematurely altering transformer attention heads, but by establishing an external, invariant-based control architecture around the reasoning engine.

2. Mathematical Formalization

2.1 The Dynamic Context State Tuple

Context is formalized not as raw conversation history, but as an actively maintained state tuple C_t :

$$C_t = \{ W_t, P_t, A_t, R_t, U_t \}$$

- **W_t (World State)**: Verified environmental state, resource limits, and external system variables.
- **P_t (Stakeholder Registry)**: Affected parties, user roles, permissions, and organizational hierarchy.

- **A_t (Assumptions & Preconditions):** Implicit and explicit premises required for action validity.
- **R_t (Relational Constraints):** Invariant rules, legal/policy boundaries, and dependency graphs.
- **U_t (Epistemic Uncertainty):** Unverified claims, ambiguous parameters, and missing information.

2.2 The Goal–Context Binding Operator

In conventional agent frameworks, an agent receives an isolated goal vector G_t . In CARA, goal validity is conditioned through an explicit binding operator:

$$B_t = f_{\theta}(G_t, C_t, V_t)$$

Where V_t denotes the set of explicit **Validity Conditions** and **Suspension Triggers**. B_t enforces that an objective is represented together with the circumstances that justify its execution.

2.3 Lexicographic Priority Hierarchy

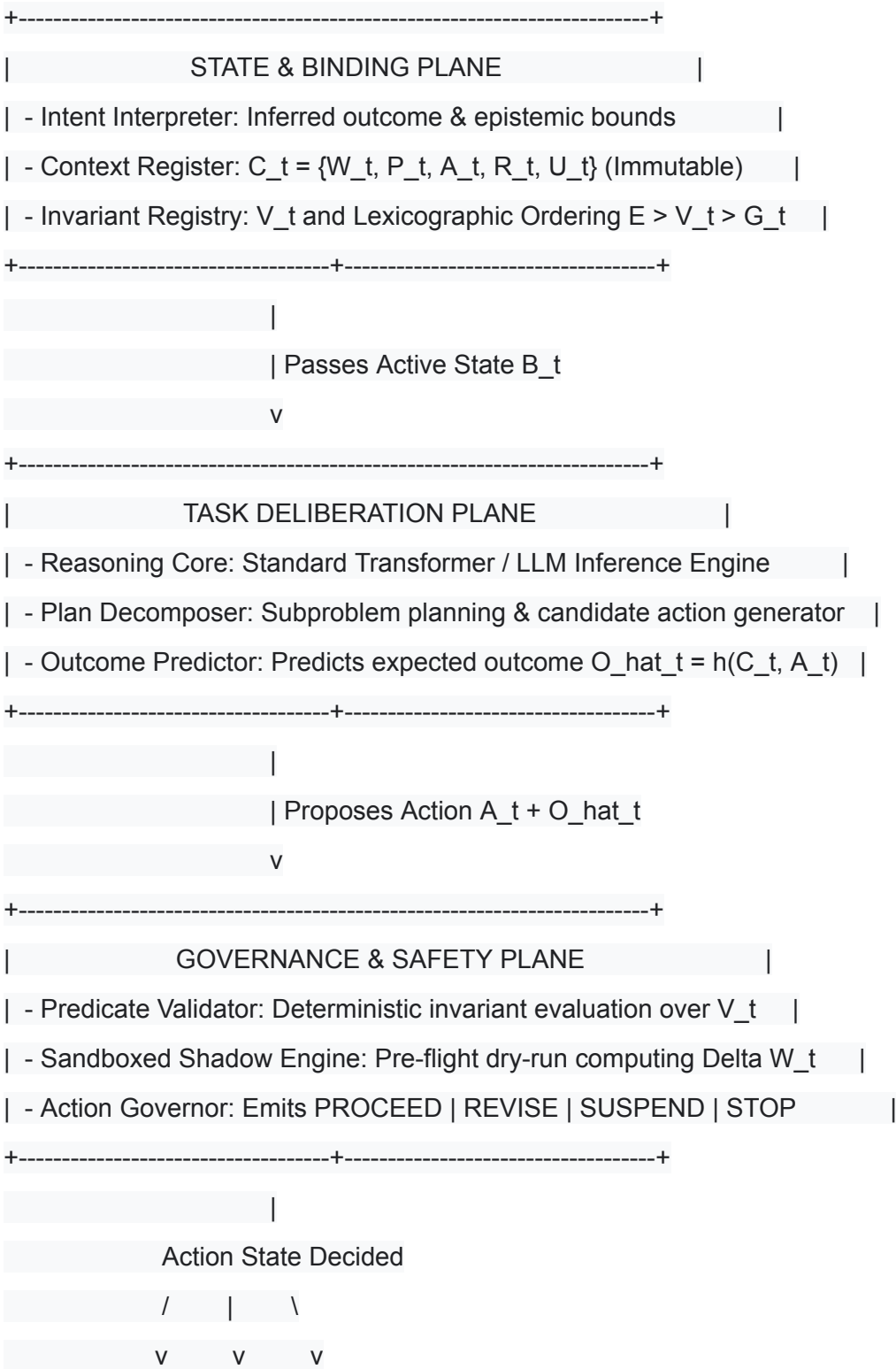
To prevent goal optimization from undermining foundational constraints, CARA establishes a strict priority ordering:

$$E > V_t > G_t$$

1. **E (Ethical & Systemic Invariants):** Universal safety constraints and harm prevention bounds.
2. **V_t (Contextual Validity Invariants):** Preconditions that make the specific task permissible in context C_t .
3. **G_t (Task Objective):** The operationalized goal requested by the user.

3. The 3-Plane System Architecture

To avoid module sprawl and serialization latency, CARA collapses control into three distinct operational planes:



PROCEED REVISE SUSPEND

(Execute) (Re-plan) (Escalate)

4. Runtime Governance & State Machine

4.1 Asynchronous Boundary Gating

Continuous token-by-token verification introduces unsustainable computational overhead and semantic noise. CARA implements **Asynchronous Boundary Gating**, evaluating invariants strictly at critical decision thresholds:

1. **Pre-Tool Emission:** Immediately before any state-mutating tool, API call, or shell execution is dispatched.
2. **Environmental Observation Arrival:** When external outputs O_t return from tools or sensors.
3. **Epistemic Divergence Spikes:** When high uncertainty or conflicting observations are detected.

4.2 Action Governor State Machine

State	Trigger Criteria	Action Taken
PROCEED	Action A_t and predicted outcome $O_{\hat{t}}$ satisfy all E and V_t . Pre-flight dry-run passes.	Action dispatches to runtime execution environment.

State	Trigger Criteria	Action Taken
REVISE	Context is valid, but proposed action violates efficiency, sub-constraints, or syntax.	Action rejected; corrective feedback fed to Task Deliberation Plane.
SUSPEND	Invalidation signal detected in V_t , unverified critical assumption broken, or ΔC_t violates purpose.	Execution immediately halted; human supervisor or clarification requested.
STOP	Objective successfully fulfilled, or environment permanently prohibits continuation.	Task terminated; summary and telemetry logged.

5. Empirical Evaluation: The Absorption Drift Benchmark

To empirically validate CARA against unconstrained baseline agents, we define a three-condition stress-testing protocol:

5.1 Test Conditions

- Condition 1 (Stable Baseline):** Standard multi-step workflow (20–40 sequential steps) with invariant context. Measures baseline task success rate and computational overhead.
- Condition 2 (Goal Invalidation Trigger):** At step $k \in [10, 25]$, introduce an

environmental shift that invalidates the objective (e.g., patient vitals drop, system compliance rule changes, database workload switches to critical transaction mode).

Evaluates **Inappropriate Continuation Rate** and **Step-to-Detection Latency**.

3. **Condition 3 (Benign Perturbation):** At step k , introduce an observable change that alters environmental state but leaves the goal validity conditions fully intact. Evaluates **False Interruption Rate**.

5.2 Evaluation Metrics Matrix

Metric	Operational Definition	Target Benchmark
Task Success Rate (TSR)	Percentage of appropriate objectives achieved in Condition 1.	Parity with unconstrained baseline ($\geq 95\%$).
Inappropriate Continuation Rate (ICR)	Frequency with which the agent persists in executing actions after invalidation (Condition 2).	$< 5\%$ (vs. $> 60\%$ in baseline agents).
Step-to-Detection Latency (SDL)	Number of actions executed between invalidation trigger and governor suspension.	1 step (immediate suspension).
False Interruption Rate (FIR)	Percentage of benign perturbations that mistakenly cause task halt (Condition 3).	$< 8\%$.

Metric	Operational Definition	Target Benchmark
Token & Latency Overhead (TLO)	Net cost of maintaining state B_t and runtime governance relative to baseline.	< 20% latency increase per action.

6. Roadmap: The Three Alignment Stages

CARA serves as the foundation for a broader three-stage alignment trajectory:

1. **Stage 1 — Reflective Intent–Context–Objective Alignment:** Cognitive and deliberative alignment via structured prompting, explicit assumptions, and reflective loops.
2. **Stage 2 — Metacognitive Transformer Architecture (MTA):** Hardware/architectural alignment via asymmetric dual-stream processing (Task Stream ↔ Context-Consistency Monitor) to prevent latent drift during raw token generation.
3. **Stage 3 — Ethical Autonomy and Governance for Superintelligence:** Normative governance for long-horizon autonomous agency, incorporating lexicographic ethical invariants, accountability protocols, and strict autonomy boundaries.

7. Conclusion

Alignment cannot remain a static condition verified only at the prompt initialization phase. As AI systems take on prolonged, autonomous execution horizons, maintaining alignment requires that goal validity be treated as a dynamic, continuous invariant. CARA decouples task reasoning from execution authority, providing a practical, testable framework to prevent absorption drift in frontier agentic systems.